

Martin J. Falk

+1 (978) 502-0180 | falkma@uchicago.edu | [linkedin.com/in/martin-j-falk/](https://www.linkedin.com/in/martin-j-falk/)

Employment

Sep 2023 – Sep 2026	Schmidt AI + Science Fellow
Oct 2019 – Sep 2023	Postdoctoral Scholar
	University of Chicago
	Advisors: Arvind Murugan and William Irvine

Education

Sep 2014 – Sep 2019	Ph.D. in Physics
	Massachusetts Institute of Technology
	Advisors: Leonid Mirny and Michael Brenner
	<i>Thesis: Self-assembly of biological heteropolymers</i>
Sep 2010 – May 2014	B.A. in Mathematics and Physics w/ honors
	University of Pennsylvania
	Advisors: Joseph Subotnik and Tom Lubensky

Preprints

- Trifonova, K., Falk, M. J., Rouches, M., Vaikuntanathan, S., Elowitz, M. B., & Murugan, A. (2025). Trainable computation in molecular networks. *bioRxiv*, 2025-12.
- Falk, M. J.*, Zhou, L.*, Matsubara, Y. J., Husain, K., Szostak, J. W., & Murugan, A. (2025). Suppression of errors in collectively coded information. *arXiv preprint arXiv:2508.21806*.

Publications

- Banerjee, D. S., Falk, M. J., Gardel, M. L., Walczak, A. M., Mora, T., & Vaikuntanathan, S. (2026). Learning via mechanosensitivity and activity in cytoskeletal networks. *Physical Review X Life*, 4(1), 013001.
- Falk, M. J.*, Strupp, A. T.*, Scellier, B., & Murugan, A. (2025). Temporal Contrastive Learning through implicit non-equilibrium memory. *Nature Communications*, 16(1), 2163.
- Falk, M. J.*, Roach, F. D.*, Gilpin, W., & Murugan, A. (2024). Curiosity-driven search for novel non-equilibrium behaviors. *Physical Review Research*, 6.3: 033052.
- Falk, M. J.*, Wu, J.*, Matthews, A., Sachdeva, V., Pashine, N., Gardel, M., Nagel, S. & Murugan, A. (2023). Learning to learn by using nonequilibrium training protocols for adaptable materials. *Proceedings of the National Academy of Sciences*, 120(27), e2219558120.
- Devany, J., Falk, M. J., Holt, L. J., Murugan, A., & Gardel, M. L. (2023). Epithelial tissue confinement inhibits cell growth and leads to volume-reducing divisions. *Developmental cell*, 58(16), 1462-1476.
- Zeng, C.*, Faaborg, M. W.*, Sherif, A., Falk, M. J., Hajian, R., Xiao, M., Hartig, K., Bar-Sinai, Y., Brenner, M. P., & Manoharan, V. N. (2022) 3D-printed machines that manipulate microscopic objects using capillary forces. *Nature* 1-6.
- Falk, M. J., Alizadehyazdi, V., Jaeger, H., & Murugan, A. (2021) Learning to control active matter. *Physical Review Research* 3.3: 033291.

- Feodorova, Y., Falk, M. J., Mirny, L. A., & Solovei, I. (2020) Viewing nuclear architecture through the eyes of nocturnal mammals. *Trends in cell biology* 30.4: 276-289.
- Falk, M. J., Duwel, A., Colwell, L. J., & Brenner, M. P. (2019) Collagen-inspired self-assembly of twisted filaments. *Physical Review Letters* 123.23: 238102.
- Falk, M. J.*, Feodorova, Y.*, Naumova, N.*, Imakaev, M., Lajoie, B.R., Leonhardt, H., Joffe, B., Dekker, J., Fudenberg, G., Solovei, I., & Mirny, L. A. (2019) Heterochromatin drives compartmentalization of inverted and conventional nuclei. *Nature* 570.7761: 395-399.
- Rocklin, D. Z., Chen, B. G. G., Falk, M. J., Vitelli, V., & Lubensky, T. C. (2016) Mechanical Weyl modes in topological Maxwell lattices. *Physical Review Letters* 116.13: 135503.
- Falk, M. J., Landry, B. R., & Subotnik, J. E. (2014) Can surface hopping sans decoherence recover Marcus theory? Understanding the role of friction in a surface hopping view of electron transfer. *The Journal of Physical Chemistry B* 118.28: 8108-8117.
- Landry, B. R., Falk, M. J., & Subotnik, J. E. (2013) Communication: The correct interpretation of surface hopping trajectories: How to calculate electronic properties. *The Journal of Chemical Physics* 139.21: 214107.

*denotes equal contribution

Awards and Fellowships

- Schmidt AI + Science Fellowship (2023-2026)
- DoD NDSEG (2015-2018)
- NSF GRFP
- University of Pennsylvania Vagelos Science Challenge Award (2012-2014)
full tuition for last two years of undergraduate degree
- Phi Beta Kappa (2013)

Mentoring experience

- Serena Debesai, Graduate Student, University of Chicago, 2025-present
- Kristina Trifonova, Graduate Student, University of Chicago, 2024-present
- Shiv Seshan, Undergraduate, University of Chicago, 2023-2025 (now Applied Physics, Caltech)
- Leon Zhou, Undergraduate, University of Chicago, 2022-2025, Goldwater Scholar (now MD/PhD, UPenn)
- Finn Roach, Undergraduate, University of Chicago, 2022-2023 (now Building Physics Analyst, Arup)
- Adam Strupp, Undergraduate, University of Chicago, 2022-2023 (now Math, UCSD)
- Ayanna Matthews, Graduate Student, University of Chicago, 2021-2023
- Jiayi Wu, Graduate Student, University of Chicago, 2021-2023

Teaching and Outreach

- Volunteer, Physics with a Bang!, Winter 2022, 2023, 2025
Volunteer for NSF MRSEC Outreach event

- Lecturer, NSF NRT-MIDAS (Computing and Data Science Training for Materials Innovation, Discovery, and Analytics), Spring 2023
Guest lecturer for “Curiosity Sampling”
- Lecturer, NAAMII Nepal AI Winter School (NepAI Applied Mathematics and Informatics Institute for research), Winter 2021
Guest lecturer for “Reinforcement Learning in Physics”
- Mentor, MIT PhysGAAP (Physics Graduate Application Assistance Program), 2020-2021
MIT initiative assisting grad school applicants from non-traditional backgrounds; mentees (2) both matriculated to Physics PhD programs at R1 universities
- Teaching Assistant, Harvard University APMTH 216, Spring 2019
Course Name: “Inverse Problems in Science and Engineering”
- Mentor, MIT PRIMES (Program for Research in Mathematics, Engineering and Science for High School Students), 2018-2019
Supervised 3 local high school students on research projects in computational biology
- Mentor, Harvard MRSEC REU, Summer 2016
Supervised 4 undergraduates on computational self-assembly project; taught core techniques of molecular simulation and thermodynamics

Funding acquisition

- Assisted in PI’s successful acquisition of Vannevar Bush Faculty Fellowship [2017-2022]
- Assisted in PI’s successful acquisition of Moore + Sloan Foundation Origin of Life Initiative [2023-]
- Assisted in PI’s successful acquisition of NSF CAREER Award [2023-]

Conferences and Talks

Contributed Talks

- “Characterizing paths between sequence-defined folds” APS March Meeting 2025, Anaheim, CA
- “Curiosity Sampling of Novel Non-Equilibrium Behaviors” APS March Meeting 2024, Minneapolis, MN
- “Contrastive Learning through Non-Equilibrium Memory” CECAM Workshop: Pathways, Memory, and Emergent Computation in Nonequilibrium Systems 2023, Ben Gurion University, Sede Boqer, IL
- “Teaching a Material to be Adaptive” American Physical Society (APS) March Meeting 2022, Chicago, IL
- “A Reinforcement Learning Approach to Optical Control of Active Matter” APS March Meeting 2021, virtual
- “Capillary Machines for Manipulating Small Objects” Division of Fluid Dynamics Annual Meeting 2020, virtual
- “Bio-Inspired Self-Assembly of Twisted Filaments”
 - o Active and Adaptive Materials Gordon Research Seminar 2019, Ventura CA
 - o Soft Matter Gordon Research Seminar 2017, New London, NH
- “Polymer Models Integrate Inverted Nuclear Geometry with Conventional HiC Compartments,” APS March Meeting 2017, New Orleans, LA

Invited Talks

- “Trainable computation in molecular systems” Systems Chemistry GRC 2026, Portland, ME
- “Non-equilibrium design principles for adaptive molecular matter” UMass Amherst Condensed Matter Seminar 2025, Amherst, MA
- “Non-equilibrium principles for adaptation in living matter” Carnegie Mellon Physics Colloquium 2025, Pittsburgh, PA
- “Design principles for adaptive molecular matter” Laufer Center Seminar 2025, Stony Brook, NY
- “Replication of Collectively Coded Information” Origins Federation Conference 2025, Chicago, IL
- “Autonomous Learning in Materials via Non-Equilibrium Memory” Complex Active and Adaptive Material Systems GRC 2025, Ventura, CA
- “Paths Between Protein Folds” Center for Living Systems Symposium 2024, Chicago, IL
- “Curiosity Sampling of Novel Non-Equilibrium Behaviors” Multidisciplinary University Research Initiative at Georgia Tech 2024, virtual
- “Teaching a Material to be Adaptive” Society for Industrial and Applied Mathematics Conference on Mathematical Aspects of Materials Science 2024, Pittsburgh, PA
- “Contrastive Learning through Non-Equilibrium Memory” Winter Conference on Computing with Physical Systems 2024, Aspen Center for Physics, Aspen, CO
- “Intrinsically-Motivated Search for Order Parameters in Non-Equilibrium Physics” FLOWERS Team at Inria University of Bordeaux 2023, virtual
- “Curiosity Sampling of Novel Non-Equilibrium Behaviors” Designing Soft Matter Lorentz Center Workshop 2023, Leiden University, Leiden, NL
- “A Reinforcement Learning Approach to Optical Control of Active Matter” Molecular Nanophotonics Group at Universität Leipzig 2021, virtual
- “Self-Assembly of Biological Heteropolymers” Materials Research Science and Engineering Center Seminar 2019, University of Chicago, Chicago, IL

Professional Activities

- Referee: Soft Matter, Nature Communications, Journal of Statistical Mechanics, Biophysical Journal, eLife, Physical Review E, Physical Review X Life
- Organizer: Computations in Science, Schmidt AI+Science Seminars

References

- Professor Margaret Gardel, University of Chicago
- Professor Jack Szostak, University of Chicago
- Professor William Irvine, University of Chicago
- Professor Arvind Murugan, University of Chicago
- Professor Vinothan Manoharan, Harvard University
- Professor Michael Brenner, Harvard University
- Professor Leonid Mirny, Massachusetts Institute of Technology
- Dr. Irina Solovei, LMU Munich